Adult cognitive function at 26, 43 and 53 years.

Adult cognitive function

With the single exception of a test repeated at age 26 years (reading comprehension), cognitive function was not assessed since adolescence until age 43 years, when there was a shift away from tests of intellectual ability and towards the assessment of functional cognitive performance (memory, processing speed, and motor praxis). Some of these tests were repeated at age 53 years, with the addition at this age of a verbal ability test (the **NART**) to link back to related tests in childhood.

Age 26 years

Age 26 years: **R26R**: the Watts-Vernon Reading Test, first given at age 15 years, was re-administered at age 26 years, with an additional 10 items of increased difficulty to avoid a ceiling effect (Rodgers, 1986).

Age 43 years

Age 43 years: Five types of cognitive test were devised by the NSHD:

1. Verbal learning. For each of three trials survey members were shown a list of 15 words at a rate of two seconds each, then were asked to write down as many words recalled as possible. A simple total score is available (**WLT89**), calculated as the sum of the words correctly recalled at each trial (**WLT891** + **WLT892** + **WLT893**).
2. Long-term recall. 1. Memory for the year (**LINTY89**), month (**LINTM89**) and day of the week (**LINTD89**) of the last interview; 2. Memory for eight medical measures (pulse, blood pressure, lung function, height, weight, and arm, chest and abdominal circumference) taken at the age 36 (1982) interview. A summary score is available: **TESTMEM** = **MEMPR89** + **MEMBP89** + **MEMLG89** + **MEMHT89** + **MEMWT89** + **MEMAC89** + **MEMCC89** + **MEMAB89**.
3. Visual memory. A 5-item delayed (20 minutes) picture recall task. A summary score is available (**PICMEM** = **PIC189** + **PIC289** + **PIC389** + **PIC489** + **PIC589**) but this has a strong ceiling effect.
4. Timed letter search. This is a cancellation task using target letters P and W, embedded among non-target letters, with 1 minute per three trials allowed, and outcome measures derived as speed, i.e. position reached at the end of each 1 minute interval, and accuracy, i.e. the number of target letters missed divided by speed. Summary scores are as follows:

CANSP89: Letter search speed: mean speed for each of the three trials, as in the following example for trial 1: CANSP891=((**VSRW189**-1)*30)+**VSCL189**, where **VSRW189** and **VSCL189** = row and column, respectively, at the end of 1 minute. A measure of letter search accuracy can be calculated by dividing the number of missed targets for each trial (**VSMS189**-**VSMS389**) by the corresponding speed score.

1. Motor speed and praxis. This task assessed time to move 10 pegs from one hole to an adjacent one. Five trials were administered for each hand. Summary scores are as follows:

PEG89: mean peg placement speed, calculated as the overall mean of five trials per hand. Separate means for the right-hand (**PGR189** to **PGR589**) and left-hand (**PGL189** to **PGL589**) trials can also be calculated. For distributional reasons a log transformation of the mean score is recommended.

Age 53 years

Age 53 years: At this age the verbal learning and timed letter search tasks used at age 43 years (nos. 1 and 4 above) were re-administered, although the latter used only one trial. In addition a delayed recall condition was added to the verbal learning task, using an interval of approximately 90 seconds. Analogous summary measures are identified by the presence of ‘99’ rather than ‘89’ in the variable names. Tests 2, 3 and 5 above were not re-administered at 53 years. However, four new tests were given at this age:

1 The National Adult Reading Test (**NART**; Nelson & Willison, 1991, variable = **NART99**). This is a pronunciation test involving 50 irregular words of increasing difficulty, chosen to violate conventional grapheme-phoneme correspondence rules. To pronounce any of the words correctly the respondent must therefore be able to recognise them in their written form rather than rely on intelligent guesswork. Thus it is effectively a test of knowledge acquisition, although it correlates with full-scale IQ. Note that the NART is traditionally scored for errors, as it is here, but this can easily be inverted by subtracting the score from 50, so as to be consistent with the direction of the other cognitive test score.

2 Verbal fluency. Category fluency was assessed by asking survey members to name as many different animals as possible in 1 minute. The score (**ANIN**) is the total number of animals named, allowing anything belonging to the animal kingdom (from amoeba to humans), but not counting repetitions, redundancies (e.g. brown cow, spotted cow…) or proper names (e.g. 'Rover', 'Kitty').

3 Prospective memory (sometimes referred to as 'remembering to remember'). Survey members were informed that, at a later stage of the interview, they would be given an envelope and asked to write a name and address on it, and that, on receipt of the envelope, they were to remember to turn it over, seal it, and write their initials on it. For the outcome variable (**REMEM**) a full score of 3 was achieved if both actions were completed without prompting; 2 if one action was achieved without prompting; and 0 if no action was undertaken without a prompt.

4 Delayed verbal memory. Survey members were asked to recall, without prior prompting, the name and address they wrote on the envelope used in the previous test ("John Brown, "). A maximum score of 6 was achievable for remembering all of these elements (NAADTA).

Variables in this document

30 high-confidence matches

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Envelope test [10134]		
remem	How well has the study member remembered tasks involved in envelope test- at age 53 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory (word list memory test) [10124]		
wlt89	Word list memory test: Sum of 3 tests (max 45) - at age 43 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Memory test [10135]		
memab89	Remembered that abdominal circumference was measured at 1982 interview - at age 43 years	bold token
memac89	Remembered that arm circumference was measured at 1982 interview - at age 43 years	bold token
membp89	Remembered that blood pressure was measured at 1982 interview - at age 43 years	bold token
memcc89	Remembered that chest circumference was measured at 1982 interview - at age 43 years	bold token
memht89	Remembered that height was measured at 1982 interview - at age 43 years	bold token
memlg89	Remembered that lung function was measured at 1982 interview - at age 43 years	bold token
mempr89	Remembered that pulse rate was measured at 1982 interview - at age 43 years	bold token
memwt89	Remembered that weight was measured at 1982 interview - at age 43 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › National audit reading test (NART) [10126]		
nart99	National Adult Reading Test (NART) pronunciation score at 53 years - naturally coded (= number of errors) - at age 53 years	bold token

Variable	Label	How it was found
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Peg placement [10122]		
peg89	Mean peg placement (both hands) - at age 43 years	bold token
pgl189	Peg-board test - first attempt - left hand (secs) - at age 43 years	bold token
pgl589	Peg-board test - 5th attempt - left hand (secs) - at age 43 years	bold token
pgr189	Peg-board test - first attempt - right hand (secs) - at age 43 years	bold token
pgr589	Peg-board test - 5th attempt - right hand (secs) - at age 43 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Processing speed [10125]		
vscl189	Visual search test - Number of columns (first test) - at age 43 years	bold token
vsms189	Visual search test - Number of misses (first test) - at age 43 years	bold token
vsms389	Visual search test - Number of misses (third test) - at age 43 years	bold token
vsrw189	Visual Search test - Number of rows (first test) - at age 43 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Reading comprehension (Watts-Vernon) [10121]		
r26r	Watts Vernon Reading Test used at age 15 (edited raw score) - at age 26 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Verbal fluency [10127]		
anin	Animal test: Number of animals listed - at age 53 years	bold token
Clinical data [51] › Cognitive function [101] › Adulthood cognition [1012] › Visual memory [10123]		
pic189	Can you remember what was on the five pictures shown earlier? First picture - at age 43 years	bold token
pic289	Can you remember what was on the five pictures shown earlier? Second picture - at age 43 years	bold token
pic389	Can you remember what was on the five pictures shown earlier? Third picture - at age 43 years	bold token
pic489	Can you remember what was on the five pictures shown earlier? Fourth picture - at age 43 years	bold token
pic589	Can you remember what was on the five pictures shown earlier? Fifth picture - at age 43 years	bold token
Questionnaire data [55] › Data collection information [545]		
lintd89	Do you remember what day of the week you were last interviewed for the National Survey? - at age 43 years	bold token
lintm89	Can you remember when you were last interviewed for the National Survey? What month? - at age 43 years	bold token
linty89	Can you remember when you were last interviewed for the National Survey? What year was that? - at age 43 years	bold token

2 medium-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Blood pressure [1103]		
timed	What is the time of day? - at 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan

3 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
hand	Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Physical capability [5108]		
arm	Do you have difficulty because of long term health problems using either arm to reach up high above your head or to reach behind to tuck a shirt in or do up a zip?- at age 53 years	prose scan